

CLINICAL TRIAL PROTOCOL

Code Name: RenalGuard-5 (CKD-203)
Study Title: A Phase 3, Multi-center, Randomized Trial for CKD Progression
Sponsor: Global Pharma Solutions Inc.
Revision: 4.0
Date: January 15, 2026

1. RATIONALE & OBJECTIVES

Chronic Kidney Disease affects 850 million individuals globally, with 1.2 million deaths annually from kidney disease complications. Current therapies only modestly slow CKD progression. RenalGuard-5 offers a novel mechanism targeting intrinsic renal pathology.

Primary Objective: To demonstrate superior reduction in proteinuria compared to placebo.

Planned Target: A mean reduction of 40% from baseline at Week 52.

2. STUDY DESIGN

This is a double-blind, randomized, placebo-controlled study across 52 sites in 12 countries. The study will enroll 900 participants aged 18-80 with CKD Stage 3-4 (eGFR 15-59 mL/min/1.73m²) and albuminuria (UACR >30 mg/g). All participants must be on stable RAAS inhibitor therapy. Study duration: 52 weeks of treatment with 8-week follow-up.

3. SAFETY MONITORING

All participants will undergo weekly monitoring for the first month, then monthly thereafter. Specific attention will be paid to renal function (serum creatinine, eGFR), electrolytes (potassium, sodium), and blood pressure.

Adverse Event Tracking: Investigators must report all serious adverse events (SAEs) within 24 hours. The Data Safety Monitoring Board (DSMB) will perform quarterly reviews of unblinded data.

Anticipated Effects: Based on Phase 2 data, expected incidence of hyperkalemia (K⁺ >5.5 mEq/L) is 6-8%, hypotension is 4-6%, and transient eGFR decline in first 4 weeks is 10-15%. No acute kidney injury grade 3 or higher was observed.

4. ENDPOINTS & STATISTICS

Primary Endpoint: Percent change from baseline in 24-hour urinary albumin excretion at Week 52.

Secondary Endpoints:

- Change in serum creatinine and eGFR
- Proportion of participants achieving 50% proteinuria reduction
- Change in blood pressure
- Time to progression to ESRD or 50% eGFR decline
- Change in markers of renal fibrosis (FGF-23, KIM-1)